Justin Yirka, PhD

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SUMMARY

Quantum computing researcher with a decade of experience. PhD from UT Austin in 2025 advised by Scott Aaronson focused on complexity theory and quantum algorithms. 9+ publications in top venues (QIP, *Quantum*). Internships at Los Alamos and Sandia National Labs. Experienced with Hamiltonian optimization problems, designing fault-tolerant quantum algorithms and NISQ algorithms, and testing algorithms with simulations (Qiskit) and real hardware (ion traps).

EDUCATION

The University of Texas at Austin
PhD in Computer Science May 2025
MS in Computer Science May 2022
Advisor: Scott Aaronson
Research Focus: Quantum computing, Algorithms, Complexity Theory

Virginia Commonwealth University
BS in Computer Science May 2018
BS in Mathematics
Minor in Physics, Certificate in Data Science. VCU Presidential Scholarship (full tuition & board).

EXPERIENCE

Blanqet | Remote August 2025 - Present
Quantum Computing Researcher
- Full-time employee #2 at a UChicago-affiliated R&D startup.
- Spearheading a novel application of quantum entanglement to a real-world industry use case.

Sandia National Laboratories | Albuquerque, NM and Remote June 2023 - May 2025
R&D Intern
- Initiated and completed a project in 6 months that was accepted to QIP (top venue).
- Analyzed and proved NP-hardness of computing product state solutions to Hamiltonian optimization problems.
- Developed new variants of Grover's (fault-tolerant) search algorithm with novel input models.

The University of Texas at Austin | Austin, TX
Graduate Research Assistant Aug 2019 - May 2025
- Secured and led a $10,000 NSF grant for a quantum seminar series, coordinating logistics and visitors.
- Developed an independent research agenda with minimal supervision, producing 6+ original research papers tackling multiple problems in quantum computational complexity.
- Identified research problems, managed multi-year projects and deadlines, and collaborated with distributed teams.
- Delivered 14+ technical presentations including at international conferences (videos available online).

Head Teaching Assistant: Quantum Information Science (online MS) Aug 2021 - May 2024
- Managed a team of up to 4 Graduate TAs and oversaw the technical instruction and grading workflows for a 200+ student advanced course for 4 semesters.
- Successfully launched the new course for the high-revenue online MS program, critical to the department's budget.

Instructor: Software Engineering with Java (online) June 2021 - July 2021
- Designed and executed a curriculum comprising hands-on Java coding projects and 30 lectures.

Los Alamos National Laboratory | Los Alamos, NM June 2019 - Aug 2019
Graduate Student Researcher
- Designed low-width NISQ algorithms for entanglement spectroscopy with error mitigation using qubit resets; published in *Quantum*.
- Validated algorithms with experiments on Honeywell (Quantinuum) ion-trap device.
- Programmed noisy circuit simulations in Python Qiskit and maintained code using git, Jupyter, and Unix tools. (link)

Graph Theory Discovery Lab at VCU | Richmond, VA May 2018 - Aug 2018
Research Assistant
- Implemented graph analysis algorithms (Python, Sage, NumPy) while improving documentation and version control.

RamDev: Software Development at VCU | Richmond, VA April 2016 - May 2018
Founder and President
- Founded and scaled to become the university's largest Computer Science organization.
- Coordinated 46 seminars and 9 corporate speakers and secured $2,400 in funding and resources.

QuICS at The University of Maryland | College Park, MD June 2017 - Aug 2017
Undergraduate Researcher
- Investigated pure-state quantum tomography with Pauli observables. Advised by Andrew Childs.

Quantum Computing Lab at VCU | Richmond, VA March 2015 - Aug 2016
Undergraduate Research Assistant
- Published 2 papers as an undergraduate, contributing key ideas for multiple proofs.
- Researched Hamiltonian complexity, characterizing the complexity of estimating measurements on low-energy states of correlated quantum systems. Advised by Sevag Gharibian.

SKILLS AND PROJECTS

Languages: Python (intermediate), Java (intermediate), C++ (beginner)
Tools: Qiskit, Sage, Mathematica, NumPy, Matplotlib, git, Unix, AWS, Agile development, LaTeX
Courses: Machine Learning, Randomized Algorithms, Natural Language Processing, Data Science, Software Engineering
Projects:
- Machine Learning course projects: PCA, ICA, Gaussian regression, Tensorflow Sept 2020 - Dec 2020
- VCU Senior Project: Bluetooth tag network with Android app, Raspberry Pi, AWS backend Sept 2017 - May 2018
  - **Award:** VCU Engineering Capstone Design Award
- Course Project for Software Engineering: Android app with geofencing, AWS backend Sept 2016 - Dec 2016
- Hackathon projects: Mathematica app with map data; Android app using REST APIs 2016

SELECTED PUBLICATIONS

Google Scholar: https://scholar.google.com/citations?user=UxIpR_UAAAAJ&hl=en
- B. Holman, R. Ramachandran, J. Yirka. Quantum search with in-place queries. TQC 2025. (link)
- J. Yirka. A note on the complexity of the spectral gap problem. arXiv:2503.02747, March 2025. (link)
- J. Yirka. Even quantum advice is unlikely to solve PP. *Theory of Computing*, 2025. (link)
- S. Grewal and J. Yirka. The Entangled Quantum Polynomial Hierarchy collapses. CCC 2024. (link)
- J. Kallaugher, O. Parekh, K. Thompson, Y. Wang, J. Yirka. Complexity classification of product state problems for local Hamiltonians. QIP 2024 and ITCS 2025. (link)
- S. Gharibian, M. Santha, J. Sikora, A. Sundaram, J. Yirka. Quantum generalizations of the Polynomial Hierarchy with applications to QMA(2). *computational complexity*, 2022. (link)
- J. Yirka and Y. Subasi. Qubit-efficient entanglement spectroscopy using qubit resets. *Quantum*, 2021. (link)
- S. Gharibian, S. Piddock, J. Yirka. Oracle complexity classes and local measurements on physical Hamiltonians. QIP 2020 and STACS 2020. (link)
- S. Gharibian and J. Yirka. The complexity of simulating local measurements on quantum systems. TQC 2017 and *Quantum*, 2019. (link)
- J. Yirka. Evaluation of TCP header fields for data overhead efficiency. NCUR 2016.
  - Awarded "Launch Award" for Outstanding Research Poster